
An Exact Conditional Null for Correlated Failure in Open-Model Ecosystems, and a Route to Anytime-Valid Monitoring

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Abstract

1 Measuring correlated failure between language models requires a null hypothesis
2 for “independent enough,” and prior work shows the conclusion is highly sensitive
3 to which null is chosen, leaving the choice apparently subjective. We show one
4 null is not a choice: the row and column margins of a model-by-item outcome
5 matrix are the jointly sufficient statistics of the Rasch family, so the uniform
6 distribution over margin-matched matrices is the unique fit-free conditional null for
7 that family. At this null, on 1,228–1,362 open models across five benchmarks, the
8 commonly reported mean-level “excess” co-failure is a deterministic function of
9 item difficulty and vanishes identically, while a second-order residual correlation
10 of $2.9\text{--}11.9\times$ its null survives, is robust to removing 21–64% of models as near-
11 duplicates, and concentrates in 4–12 latent dimensions rather than one shared factor.
12 Our test is a fixed-sample exact randomization test. Because open leaderboards
13 grow continuously, the natural downstream use is a standing monitor, which is
14 an anytime-valid inference problem. We state what the exact null already gives
15 for that purpose (a valid per-look p -value, hence calibrated e -values that merge
16 validly across a fixed set of looks) and what it does not (an e -process with optional
17 stopping), and pose the missing construction as an open problem for this workshop
18 rather than as a result.

19 1 Motivation

20 If many deployed models share blind spots, an ensemble, a panel of model judges, or a market of
21 competing screening vendors supplies far less independent evidence than its headcount suggests.
22 A growing literature measures how often models err together (Kim et al., 2025), almost all of it
23 by comparing the rate at which models fail the same items against a baseline in which failures are
24 independent. Jo et al. (2026) show this comparison is highly sensitive to the assumed baseline:
25 conditioning on item difficulty attenuates residual correlations, “some even flipping from strongly
26 positive to slightly negative,” and they prove that a sufficiently expressive null can explain away any
27 observed dependence. Their stated conclusion is that the choice of null is subjective. This paper
28 argues that the ladder of nulls has a canonical rung, reports what the exact answer is there, and states
29 precisely what that answer does and does not provide for the sequential-monitoring problem this
30 workshop’s machinery addresses.

31 2 The canonical null

32 Let $F \in \{0, 1\}^{N \times M}$ record $F_{im} = 1$ iff model i fails item m , with row sums r_i and column
33 sums c_m . The Rasch model $\Pr(F_{im} = 1) = \sigma(\alpha_i + \beta_m)$ has $\{r_i\}$ and $\{c_m\}$ as jointly sufficient

Benchmark	N	rms $ R_{ij} $			participation ratio			dims > edge
		obs.	null	ratio	obs.	null	ratio	
ARC-Challenge	1362	0.2483	0.0449	$5.5\times$	23.6	449.3 ± 3.4	0.052	6
Winogrande	1361	0.2691	0.0409	$6.6\times$	20.1	493.8 ± 3.0	0.041	5
TruthfulQA	1334	0.2942	0.0583	$5.1\times$	17.4	309.7 ± 2.4	0.056	5
GSM8K	1228	0.0987	0.0346	$2.9\times$	118.6	553.1 ± 2.2	0.214	4
HellaSwag	1362	0.2457	0.0206	$11.9\times$	27.3	958.0 ± 3.2	0.028	12

Table 1: Residual structure after exact-margin conditioning. Null rms is the mean over 40 curveball replicates; null PR is mean $\pm$ SD over 60 independent-chain replicates, each burned in for 50 trades per model with both margins exactly preserved. Uncalibrated, PR reads 1.5–3.6 on every benchmark, a number driven almost entirely by the shared item-difficulty eigenvalue that conditioning removes.

statistics, so the conditional law of F given its margins is uniform on the set of margin-matched binary matrices for every parameter value (Rasch, 1960; Andersen, 1973). That set is the ecologists’ fixed-fixed null. We sample it with curveball (Strona et al., 2014), whose convergence to the uniform distribution on the fibre is proved by Carstens (2015) provided failed trades are counted as steps (ours does), and which we verify directly: enumerating every matrix on four small fibres (sizes 48, 90, 204, 1,170) and drawing 60,000 thinned samples, a χ^2 test against uniformity is not rejected on any ($p = 0.23, 0.05, 0.73, 0.15$); on the real matrices, four dispersed chains give Gelman–Rubin $\hat{R} \in [0.98, 1.03]$ at the burn-in used. Unlike a fitted item-response null, this null requires no optimizer, no penalty, and no identification convention: within the Rasch family it is the strongest ability-and-difficulty control obtainable without estimating anything.

Lemma 1 (Mean co-failure depends only on item margins). *The mean pairwise co-failure rate satisfies $O = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{M} \sum_m F_{im} F_{jm} = \frac{1}{MN(N-1)} \sum_m (c_m^2 - c_m)$.*

Proof. $\sum_{i \neq j} F_{im} F_{jm} = (\sum_i F_{im})^2 - \sum_i F_{im}^2 = c_m^2 - c_m$ since $F_{im} \in \{0, 1\}$. $\square$

Any randomization preserving the item margins therefore leaves O invariant with exactly zero variance, so the “excess” of O over this null is identically zero by construction rather than empirically small. The same degeneracy is known in ecology as that of Schluter’s V-ratio (Schluter, 1984; Gotelli, 2000), in psychometrics as Cronbach’s α (Cronbach, 1951), and in inter-rater agreement as Fleiss’s observed agreement (Fleiss, 1971); to our knowledge none of these is cited in the model-evaluation literature that reports the statistic, and the full paper states and attributes the identities.

3 What survives at the exact null

Statistics. Let $P_{im} = \sigma(\hat{\alpha}_i + \hat{\beta}_m)$ be the maximum-entropy (Rasch) mean field matching the observed margins, fitted to a maximum absolute margin error below 10^{-11} . It is a point estimate of the null mean of $FF^\top/M$, and we validate it against the randomization itself: over 319,600 model pairs on ARC, the curveball expectation of $C_{ij} = \frac{1}{M} \sum_m F_{im} F_{jm}$ and the Rasch prediction correlate at 0.99999. The residual matrix is $D = (FF^\top - PP^\top)/M$ and R is D scaled to unit diagonal. We report (i) the root-mean-square off-diagonal $|R_{ij}|$ and (ii) the participation ratio $\text{PR} = (\sum_k \lambda_k^+)^2 / \sum_k (\lambda_k^+)^2$ of R ’s positive eigenvalues, each against the same quantity on exact-margin curveball replicates, and (iii) the number of eigenvalues of R exceeding the null’s largest eigenvalue (parallel analysis against the exact null rather than against iid noise).

Data. Per-item outcomes are harvested from the archived Open LLM Leaderboard (1,228–1,362 models per benchmark after removing degenerate rows and columns; five benchmarks) at zero inference cost by reading only the metric column of each result file. Item identity is resolved by row index, licensed by a verified zero-mismatch row-order audit over 334 models spanning the archive’s two schema generations.

Results. Table 1 shows residual correlation at 2.9 – $11.9\times$ its null and a participation ratio at 2.8 – 21.4% of its null on every benchmark. Neither number licenses the reading “1,362 models behave

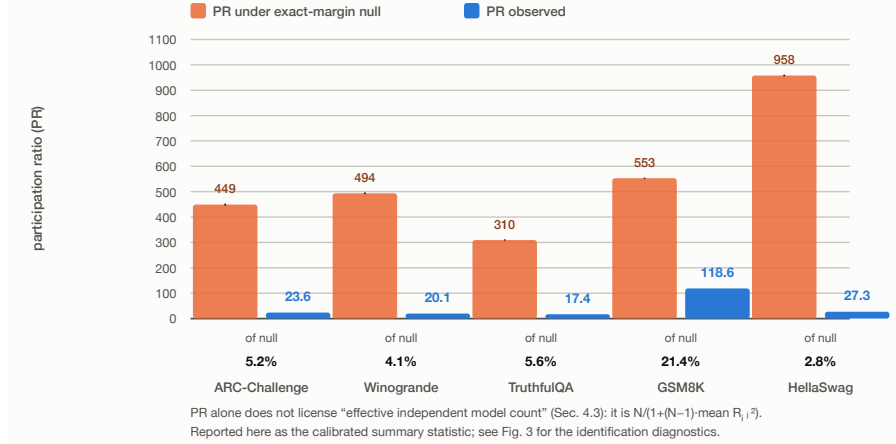


Figure 1: Observed participation ratio against its exact-margin null on all five benchmarks. Reported as a calibrated summary statistic, not as a count of independent models (Section 3).

like 24”: for the unclipped spectrum, $PR = N/(1 + (N - 1) \overline{R_{ij}^2})$ exactly (16.04 on ARC computed either way), so PR is a monotone transform of mean squared residual correlation and cannot separate one weak global factor from a partition into tight clusters. The diagnostics that can separate them say the structure is neither: on ARC the leading eigenvalue carries 54% of squared residual mass (a single factor would carry $\approx 99\%$, twenty exact clone blocks $\approx 10\%$), deflating it raises PR only to 46.8 rather than toward N , and six eigenvalues exceed the null’s spectral edge of 27.9. A two-parameter item-response simulation with discrimination heterogeneity of $sd(\log a) = 0.60$ produces rms residual correlation of 0.076 against the observed 0.248, so discrimination heterogeneity alone is not a sufficient explanation, while a simulated population with two ability dimensions and no shared lineage reproduces much of the effect ($\lambda_1^2 / \sum \lambda_k^2 = 0.457$ against the observed 0.540).

Robustness. Recomputing every statistic on populations deduplicated at agreement threshold 0.95, which discards 21–64% of models, with the Rasch fit and the null redrawn each time, moves the rms ratio by at most $1.14\times$ (ARC $5.58 \rightarrow 5.44$) and leaves the eigenvalue count unchanged on three of five benchmarks; the PR ratio is the one statistic that moves materially, by $2.44\times$ on HellaSwag and by at most $1.46\times$ elsewhere. A later pass of the identical estimator over the growing archive (2,972–3,762 models per benchmark, $2.2\text{--}2.8\times$ the population here) moved every PR ratio down, by 0.005–0.010 on four benchmarks and by 0.023 on GSM8K. On the independently implemented successor harness (ARC, 212 models) the rms excess is $4.1\times$ its null with four eigenvalues above the edge, agreeing in direction but not in magnitude. On planted data (400 models, 700 items, 20 datasets per condition), the test reaches power 1.00 against shared failure modes at a planted rms of 0.048 and rejects at 0.017 under the null; against the alternative in which every model shares one item-difficulty profile and differs only in ability, power is exactly α , because that alternative is the null by construction.

A refuted pre-registration. The pre-registered primary statistic was the dispersion $\text{Var}_{i < j}(C_{ij})$. It came out below its null on three benchmarks and above on two, its kill condition fired, and we report it as refuted rather than reinterpreted. The second-order structure above is what survives, not the statistic we planned to report.

4 What the exact null gives, and does not give, for anytime-valid monitoring

Everything above is a fixed- N exact test. The population is not fixed: the leaderboard and its successors add models continuously, and the natural downstream use of the test is a standing monitor (“has correlated failure in the open ecosystem worsened since last quarter?”) checked repeatedly as submissions arrive, without the type-I inflation that repeated testing on a growing sample otherwise incurs. That is the problem this community formalizes with e-values and test martingales (Shafer, 2021; Vovk & Wang, 2021; Ramdas et al., 2023; Grünwald et al., 2024).

104 **What is already in hand.** At any population size N_t the null law of the accumulated matrix given
 105 its current margins is exactly uniform on a finite set, so the Monte Carlo p -value p_t at each look
 106 is exactly valid. Two consequences follow directly from the e-value literature. Any single p_t can
 107 be calibrated into an e-value, for instance $\kappa p_t^{\kappa-1}$ with $\kappa \in (0, 1)$ (Vovk & Wang, 2021); and the
 108 arithmetic mean of e-values is an e-value under arbitrary dependence (Vovk & Wang, 2021), so
 109 averaging calibrated e-values over a pre-specified set of monitoring dates is a valid combined test
 110 across those looks even though nested populations are strongly dependent. What this does not give
 111 is optional stopping: it is valid for a fixed set of looks, not for stopping the first time the running
 112 evidence looks large, which is the guarantee an e-process provides (Ramdas et al., 2023).

113 **What is missing, and why.** A test martingale needs a null for the transition from the population
 114 at time t to that at $t + 1$. Conditioning on the margins of the accumulated matrix eliminates the
 115 Rasch parameters at each fixed t , but the margins themselves change when a row is added, and
 116 the conditional argument does not supply the null distribution of that increment: given the first N_t
 117 rows and the new margins, the new row is determined. The sequential object must therefore be built
 118 differently, for instance by conditioning each arriving row on its own row sum (which eliminates
 119 that model’s ability parameter but not the item parameters) and handling the item parameters as a
 120 composite nuisance with a mixture or plug-in e-value over the evidence accumulated about them, at
 121 the cost of the exactness that motivates the fixed- N test. A second obstacle is selection: new models
 122 are not exchangeable arrivals but are submitted by people who observe the current leaderboard,
 123 which is the population-dependence objection Jo et al. (2026) raise about a fixed null, now posed
 124 sequentially. We state both as the concrete open problem we would bring to this workshop, not as a
 125 result. The exact conditional null is, we think, the right fixed-sample primitive for an anytime-valid
 126 ecosystem monitor; we have not built the monitor.

127 5 Limitations

128 Claims cover the evaluated open-model ecosystem on multiple-choice-style benchmarks, not deployed
 129 proprietary systems, and the population is self-selected leaderboard submissions. The conditioning
 130 model is unidimensional; Section 3 rules out discrimination heterogeneity and duplicate models as
 131 sufficient explanations but cannot separate genuine behavioural concentration from an underfit null
 132 family, and on the spectral evidence the latter is the better description. Two candid points about the
 133 null itself: its rms sits only $1.25\text{--}2.0\times$ above the analytic $1/\sqrt{M}$ floor, and conditioning on the item
 134 margins alone reproduces it to within 8.3% on every benchmark, so the item margins do essentially all
 135 of the conditioning work for this statistic. Code, the harmonised substrate, and every results artifact
 136 are archived at doi:10.5281/zenodo.21860005 (<https://github.com/SilverCoin256/cofail>);
 137 the pre-registration with dated amendments is at <https://osf.io/cmh7q/>. A large language
 138 model (Claude, Anthropic) was used as a coding and drafting assistant; the author specified and pre-
 139 registered the hypotheses, verified every derivation and every reported number against executed-run
 140 artifacts, and is responsible for the content.

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